

Lab exercise 8

Query with Sub Query& Correlated Query

AIM:

To perform subquery and correlated query on the given relations.

DESCRIPTION:

SUBQUERY

A MySQL subquery is a query nested within another query such as SELECT, INSERT, UPDATE or DELETE. In

addition, a MySQL subquery can be nested inside another subquery.

A MySQL subquery is called an inner query while the query that contains the subquery is called an outer query. A

subquery can be used anywhere that expression is used and must be closed in parentheses.

1. Which vegetable has a price greater than the average price?

Subquery:

```
SELECT VegName, Price
FROM Vegetable
WHERE Price > (SELECT AVG(Price) FROM Vegetable);
```

Output:

VegName	Price
Cabbage	60.00
Beans	63.00
Capsicum	67.00

2. Which vegetables belong to a category having more than one vegetable?

Correlated Subquery:

```
SELECT V1.VegName, V1.Category
FROM Vegetable V1
WHERE (SELECT COUNT(*)
      FROM Vegetable V2
      WHERE V2.Category = V1.Category) > 1;
```

OUTPUT:

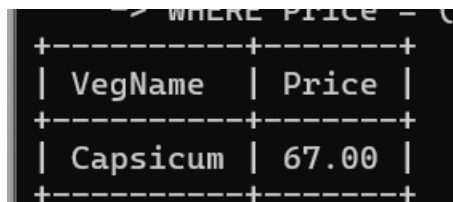
VegName	Category
Tomato	Vegetable
Potato	Vegetable
Onion	Vegetable
Brinjal	Vegetable
Cabbage	Vegetable
Beans	Vegetable
Capsicum	Vegetable

3. Which vegetable has the highest price?

Since your table does **not have a date/joined-date column**, the original faculty question cannot be directly applied.

A suitable **subquery** for your table is:

```
SELECT VegName, Price
FROM Vegetable
WHERE Price = (SELECT MAX(Price) FROM Vegetable);
```



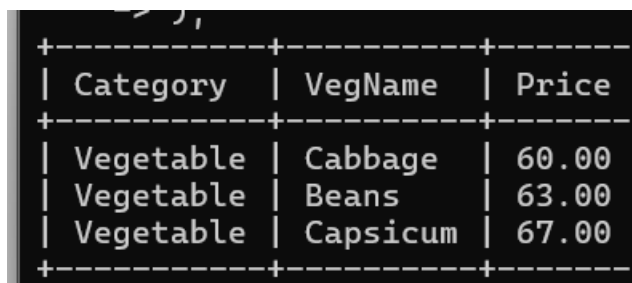
```
SQL> WHERE PRICE = (SELECT MAX(PRICE) FROM VEGETABLE);
```

VegName	Price
Capsicum	67.00

4. List the category and price of vegetables whose price is greater than the average price of their category.

Correlated Subquery:

```
SELECT V1.Category, V1.VegName, V1.Price
FROM Vegetable V1
WHERE V1.Price > (
    SELECT AVG(V2.Price)
    FROM Vegetable V2
    WHERE V2.Category = V1.Category
);
```



```
SQL> ,
```

Category	VegName	Price
Vegetable	Cabbage	60.00
Vegetable	Beans	63.00
Vegetable	Capsicum	67.00